

ServiceNow GlideForm (g_form) Overview

Complete Client-Side Form Manipulation Reference Manual

The GlideForm API (g_form) represents the client-side form instance. It allows developers to customize form behaviour, modify field parameters, perform asynchronous table reads, and supply instant inline messages to improve the user experience.

1. Core Specifications

- **Purpose:** Main API class used to control fields, validate forms, and format layouts dynamically on the client side.
- **Scope:** Only available in environments that run client scripts, UI policies (scripts), and client-callable UI Actions.
- **Key Use Cases:** Changing field properties (mandatory, read-only, visible), retrieving or setting field values, modifying labels, displaying banner/field errors, and performing lazy-loading queries.

2. Commonly Used Methods (Values & Visibility)

A. Value Manipulation

`g_form.getValue(fieldName)`

Fetches the current form input value as a string. Note that for checkboxes, it returns string 'true' or 'false'.

```
var urgency = g_form.getValue('urgency');
if (urgency === '1') {
    // Execute priority logic
}
```

`g_form.setValue(fieldName, value, displayValue)`

Sets the value of a field. For Reference fields, providing a third 'displayValue' parameter avoids unnecessary, slow AJAX queries to fetch the displays.

```
g_form.setValue('short_description', 'Urgent server patch required');
// Reference field optimization:
g_form.setValue('assigned_to', '5137153ec611227c0000851e3e7f2235', 'John Doe');
```

`g_form.clearValue(fieldName)`

Clears the value of a field entirely.

```
if (g_form.getValue('u_hardware_type') === '') {  
    g_form.clearValue('serial_number');  
}
```

B. Visibility & Display

`g_form.setDisplay(fieldName, true/false)`

Hides or displays the field. When set to false, the field and its label are hidden, and the form layout collapses to remove any blank gap (Standard Best Practice).

```
g_form.setDisplay('work_notes', false); // Clean layout collapse
```

`g_form.setVisible(fieldName, true/false)`

Hides or displays the field. When set to false, it hides the field but keeps the blank placeholder space on the page (not recommended for clean layouts).

```
g_form.setVisible('configuration_item', false); // Leaves empty visual gap
```

3. Commonly Used Methods (Properties & Messages)

A. Mandatory & Read-Only Flags

`g_form.setMandatory(fieldName, true/false)`

Defines whether the field must contain a value prior to form submission.

```
g_form.setMandatory('close_notes', true);
```

`g_form.setReadOnly(fieldName, true/false)`

Prevents the user from modifying the value of the specified field.

```
g_form.setReadOnly('opened_by', true);
```

B. Labels & Feedback Messages

`g_form.setLabel(fieldName, 'New Label')`

Changes the visible text label of the field dynamically.

```
g_form.setLabel('u_alternate_contact', 'Emergency Phone');
```

```
g_form.showFieldMsg(fieldName, 'Message', 'type')
```

Displays a styled inline validation message below the target field. 'type' can be 'info', 'error', or 'warning'.

```
g_form.showFieldMsg('u_email', 'Please double-check your email domain', 'warning');
```

```
g_form.hideFieldMsg(fieldName)
```

Removes the inline validation message currently shown under the target field.

```
g_form.hideFieldMsg('u_email');
```

4. Reference Fields & Utility APIs

A. Reference Fields (Asynchronous Retrieval)

```
g_form.getReference(fieldName, callback)
```

Retrieves the full database record for a referenced item. You **MUST** supply a callback function. Without a callback, the call is synchronous, freezing the browser and triggering performance alerts.

```
g_form.getReference('caller_id', function(userRecord) {  
    g_form.setValue('u_phone', userRecord.phone);  
    g_form.setValue('u_email', userRecord.email);  
});
```

B. Utility APIs

- **g_form.getFieldNames():** Returns a JavaScript array of all field names present on the form.
- **g_form.isMandatory(fieldName):** Checks if the field is currently required before submit (returns true/false).
- **g_form.isVisible(fieldName):** Checks if the field is currently visible to the user (returns true/false).
- **g_form.isReadOnly(fieldName):** Checks if the field is currently locked/read-only (returns true/false).

5. Best Practices & Guidelines

- [x] Always enclose field names in quotes (e.g. `g_form.getValue('short_description')`). Passing unquoted variable tokens causes JavaScript syntax errors.
- [x] Always use a callback function with `getReference()`. Synchronous database lookups block the browser main thread and violate ServiceNow design certification rules.
- [x] Prefer `setDisplay()` over `setVisible()` to ensure hidden fields do not leave empty gaps, keeping the form layout clean.
- [x] Use JavaScript native `Date()` class properties for client-side date operations. Server-side classes like `GlideDateTime` cannot run client-side and will throw errors.